

QUESTION-TO-SLIDE MAPPING

SECTION B - LECTURE 03.1

Elements of Climate for Building Designers: Driving Rain, Solar, Wind & Sky

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	3.1 - Elements of Climate for Building Designers (11 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Mapping rule	Treat 03.1 as its own lecture, separate from 03.0. Map the developed topics only: Driving Rain/DRI, sky condition, solar-radiation duration vs intensity, and wind measurement/wind rose. Vegetation is named on the title slide but not developed in the provided PDF, so it is not treated as full coverage.

RESULT

Lecture 03.1 contains one of the clearest modern-paper hot zones: Driving Rain / DRI appears in four consecutive papers (2021-2024). It also supports the repeated wind-frequency / wind-rose questions and the airflow-measurement part of 2024. Solar measurement and sky-condition data are useful supporting topics, but some global-radiation questions still require Lecture 02.

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

1. Executive Summary

This 11-slide deck continues the Elements-of-Climate topic from a building-designer viewpoint. It defines wind-driven rain, gives the DRI formula and exposure zones, explains sky-condition/cloud-cover recording, distinguishes sunshine duration from radiation intensity and lists instruments, then covers wind speed/direction measurement, anemographs and wind-rose interpretation. The historical question bank strongly rewards the DRI and wind blocks.

11	10	4	1	5	102
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Highest-priority slide blocks

Slides	Lecture block	Historical signals	Exam value
2-4	Driving Rain + DRI formula + sheltered/moderate/severe exposure	2021, 2022, 2023, 2024	Highest priority; four consecutive-paper family.
8-11	Wind instruments, recording heights, wind rose / frequency interpretation	2017 Q3(b); 2024 Q4(a), Q7(c)	High return; repeated graph/measurement family.
6-7	Sunshine duration vs radiation intensity + sunshine recorder	2021 solar quantity partial; 2024 radiation support	Useful measurement block; global solar constant/surface-radiation mechanism is in Lecture 02.
5	Sky condition / cloud cover observations	Supports climatic-data questions	Short-note / supporting data block.
1	Topic map incl. vegetation	2024 climatic-factor question partial	Vegetation is named but not developed in this PDF.

Priority note Prepare Slides 2-4 first: formula + exposure classification + building implication. Then practise one wind rose from Slides 10-11 and memorize the instrument/height rules on Slide 9.

2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2017	Q3(b)	20	2	Draw monthly and annual wind-frequency graphs and interpret them.	Partial	8-11	Wind measurement and wind-rose interpretation are shown, but the lecture does not explicitly teach the monthly-versus-annual plotting procedure.
2021	Q3(c)(i)	5	6	Write short note on solar radiation quantity.	Partial	6-7	The deck explains duration versus intensity and instruments; solar constant and global quantity variation are in Lecture 02.
2021	Q3(c)(iii)	5	6	Write short note on driving rain.	Exact	2-4	Definition, DRI formula, exposure meaning and zones are directly supplied.
2022	Q3(c)	10	8	Derive driving rain. With its detail how to design the exposure of a building using driving rain index.	Exact	2-4	The formula, interpretation and sheltered/moderate/severe exposure classes directly answer the question.
2023	Q2(c)	10	10	Describe driving rain and Driving Rain Index.	Exact	2-4	Direct repeated DRI match.

2. Detailed Year-wise Question Mapping - continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2024	Q2(a)	7	13	Define driving rain. Write in detail how to design building exposure using driving rain index.	Exact	2-4	Direct match to DRI definition, formula and exposure classification.
2024	Q4(a)	13	13	Draw a monthly and annual wind-frequency graph and explain how they are usually interpreted.	Direct	8-11	Wind rose direction, frequency, speed classes and interpretation are shown; the student must adapt the same logic to monthly/annual data.
2024	Q4(b)	13	13	What climatic factors affect building design and how do they influence it?	Partial	1-11	This deck covers rain, solar, wind and sky, but not temperature/humidity and it only names vegetation.
2024	Q4(c)	9	13	Evaluate radiation of the Earth surface from the viewpoint of global climate variation.	Partial	6-7	Radiation intensity/duration measurement is present, but incidence, atmospheric depletion and Earth-surface balance belong to Lecture 02.
2024	Q7(c)	10	15	Describe measuring temperature, humidity and air flow at a specific site.	Partial	8-9	Airflow speed/direction instruments and field heights are covered; temperature and humidity measurement are in Lecture 03.0.

Coverage key

Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required.

3. Repeated Question Patterns

The exam value is concentrated rather than broad. DRI is a four-year consecutive repeat, while wind-frequency interpretation returns after a long gap. This means Slides 2-4 and 8-11 deserve disproportionate revision time.

P	Normalized repeated family	Years seen	Slides	Why it matters
S	Driving rain + DRI + exposure classification	2021, 2022, 2023, 2024	2-4	Four consecutive papers; formula + design implication.
A	Monthly/annual wind-frequency graph / wind rose interpretation	2017, 2024	8-11	High-mark graph question; understand direction/frequency/speed.
A	Wind / airflow measurement	2024 (+ general support)	8-9	Instrument + recording-height facts are exam-ready.
B	Solar radiation quantity / duration vs intensity	2021, 2024 related	6-7	Useful measurement content; global solar mechanism remains in Lecture 02.
B	Sky condition / cloud cover	Support to climatic-information questions	5	Short, easy supporting topic.

4. Quick Recognition Rules

If the question says...	Go to slides	Recognition rule
"driving rain / DRI / building exposure"	2-4	DRI = annual rainfall (m) x annual average wind speed (m/s); classify sheltered <=3, moderate 3-7, severe >7 m²/s.
"wind frequency / wind rose"	10-11	Spoke direction = where wind comes from; length = frequency; colour/segment = speed class; centre may show calm.
"measure wind / airflow"	8-9	Speed: cup/propeller anemometer; direction: wind vane; continuous: anemograph; open terrain ~10 m.
"solar duration vs intensity"	6-7	Duration = hours/day from sunshine recorder; intensity = power/area from radiation instruments.
"sky condition"	5	Cloud presence/cover in tenths/eighths; observation time/frequency matters.
"temperature/humidity measurement"	Lecture 03.0	Do not mix the reports; those instruments are taught in the previous deck.

5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

Slides	What the slides teach	Past-question links	Study action
1	Topic map: driving rain, solar, wind, sky, vegetation	2024 Q4(b) partial	Use only as a checklist; vegetation is not developed.
2-4	Driving rain, DRI formula, example and exposure zones	2021 Q3(c)(iii); 2022 Q3(c); 2023 Q2(c); 2024 Q2(a)	Memorize formula + thresholds + one design response per zone.
5	Sky condition/cloud-cover recording	Climatic-information support	Prepare as a 3-5 mark short note.
6-7	Sunshine duration vs radiation intensity; sunshine recorder	2021 Q3(c)(i) partial; 2024 Q4(c) partial	Know the distinction and instruments; open Lecture 02 for global radiation.
8-9	Wind instruments, field heights and units	2024 Q7(c); supports graph questions	Memorize instrument -> parameter -> height.
10-11	Wind rose and interpretation	2017 Q3(b); 2024 Q4(a)	Practise a labelled wind rose and a 4-sentence interpretation.

6. Coverage Gaps & Diagnostic Exclusions

This deck is strong on DRI and wind, but it does not by itself complete every “Elements of Climate” question. The biggest mistakes would be using it alone for temperature/humidity measurement or for global radiation at the Earth surface.

Gap	Affected questions	Action
Temperature and humidity measurement are absent	2024 Q7(c)	Use Lecture 03.0 Slides 4-7, then combine with 03.1 wind slides.
Global solar constant / Earth-surface radiation mechanism is absent	2021 solar quantity; 2024 Q4(c)	Use Lecture 02 Slides 8 and 11-13 for the global radiation answer.
Monthly vs annual wind-graph construction is not explicitly shown	2017 Q3(b); 2024 Q4(a)	The wind rose interpretation is direct; use class-note examples for exact monthly/annual plotting if supplied.
Vegetation is named but not explained	2024 Q4(b)	Use the Section B class note/reference for vegetation effects; do not invent them from Slide 1.
Temperature/humidity/precipitation fundamentals are in the separate 03.0 deck	Broad climatic-information questions	Keep the two lecture reports paired but separate.

Mapping discipline

The DRI thresholds are taken from the uploaded 03.1 deck exactly as shown: Sheltered $\leq 3 \text{ m}^2/\text{s}$; Moderate 3-7 m^2/s ; Severe $> 7 \text{ m}^2/\text{s}$. The report also preserves the deck distinction between sunshine duration and radiation intensity.

7. Recommended Study Plan

This is a high-efficiency lecture: DRI can be mastered quickly and has repeated four years in a row. After DRI, spend time on a wind rose and the instrument rules; only then review sky/solar short notes.

Step	Study block	Slides	Practice immediately
1	Driving Rain + DRI	2-4	2022 Q3(c), 2023 Q2(c), 2024 Q2(a).
2	Wind rose / frequency interpretation	10-11	2017 Q3(b), 2024 Q4(a).
3	Wind measurement / airflow	8-9	2024 Q7(c) airflow part.
4	Solar duration vs intensity	6-7	2021 solar quantity, then compare with Lecture 02.
5	Sky condition	5	Prepare a concise short note + why observation time matters.

Exam-ready minimum

Priority	What to reproduce	Exam technique
A	DRI formula + thresholds	Always state unit m ² /s and finish with the exposure class + envelope implication.
A	Wind rose interpretation	Direction FROM which wind comes; spoke length = frequency; segments/colour = speed.
A	Wind instruments	Anemometer = speed; vane = direction; anemograph = continuous record.
B	Duration vs intensity	Do not write them as the same quantity; give separate units/instruments.

Bottom line

Lecture 03.1 is a modern-paper scoring lecture. DRI is the must-prepare item; wind rose and airflow measurement are the next best returns. Keep global radiation and temperature/humidity measurement linked to - but sourced from - Lectures 02 and 03.0.